Multi-Region Disaster Recovery Plan

Services: EC2, RDS

Description:

Implement a disaster recovery architecture where applications and databases are deployed across two AWS regions. Configure cross-region database replication to ensure business continuity and data availability during regional outages.

Skills: High Availability, Disaster Recovery Planning, Cross-Region Replication, Database Management, Business Continuity.

Definition: The Multi-Region Disaster Recovery Plan is designed to provide a resilient and highly available infrastructure by maintaining application and database resources in two separate AWS regions. The primary region hosts the production workload, while the secondary region acts as a disaster recovery (DR) environment.

The solution uses Amazon EC2 instances for application hosting and Amazon RDS cross-region replication for database synchronization. In the event of a failure in the primary region, services can be restored from the secondary region, minimizing downtime and reducing data loss.

Scope of the Project:

The scope of this project is to design and implement a disaster recovery architecture using AWS services. The project focuses on ensuring application availability and database protection through cross-region deployment and replication.

This project covers:

- Primary Region Deployment: Deploy application servers and databases in the primary AWS region.
- Disaster Recovery Region Deployment: Deploy standby infrastructure in a secondary AWS region.
- Cross-Region Database Replication: Replicate data from the primary RDS database to a secondary region using Read Replicas.
- Disaster Recovery Testing: Simulate failures and validate recovery procedures.
- Business Continuity: Ensure applications remain available during regional outages.

Methodology:

1**. Cross-Region Replication Method – Best Method**

- Uses AWS services: EC2 and RDS.
- Provides continuous database replication between regions.
- Reduces Recovery Point Objective (RPO).
- Suitable for production and enterprise workloads.
- Ensures business continuity during disasters.
- Requires database replication configuration and monitoring.

2. **Backup and Restore Method – Alternative Method**

- Uses RDS automated backups and snapshots.
- Lower infrastructure cost.
- Simpler implementation.
- Suitable for development and testing environments.
- Recovery process requires manual database restoration.
- Recovery time is higher compared to replication-based DR.

AWS Services used in the Project

1. **Amazon EC2 (Elastic Compute Cloud)**

Definition: Amazon EC2 is a web service that provides secure and resizable virtual servers in the cloud, allowing users to deploy applications and workloads on demand.

Purpose: Provides application servers in both primary and disaster recovery regions.

Role: Hosts web applications and services that user's access.

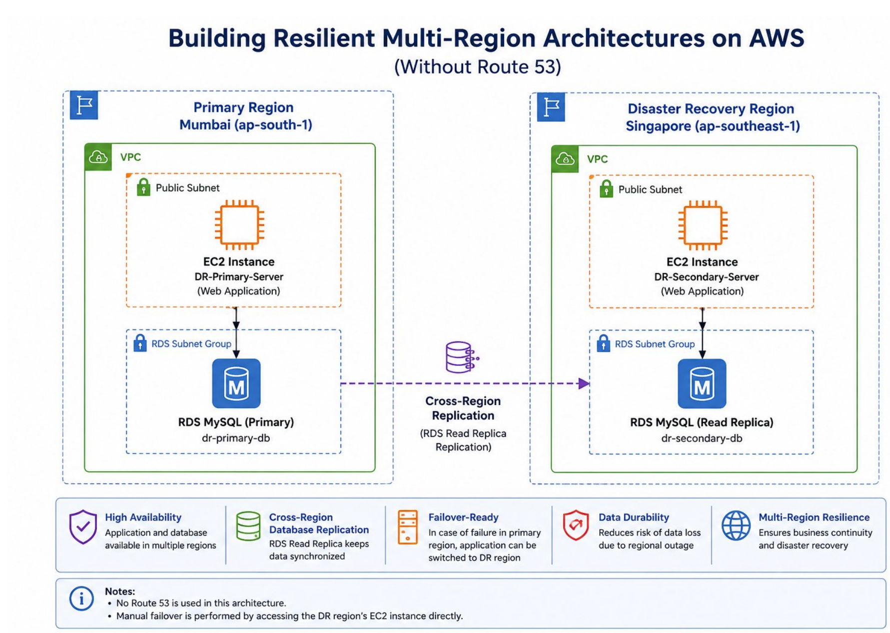
2. Amazon RDS (Relational Database Service)

Definition: Amazon RDS is a managed database service that simplifies database administration, backups, patching, monitoring, and replication.

Purpose: Stores application data and provides database replication capabilities.

Role: Maintains the primary database and replicates data to the disaster recovery region through Read Replicas.

Architecture Diagram



Implementation Steps to Do the Project

Step 1: Launch Primary EC2 Instance

• Go to AWS Console → EC2 → Launch Instances.

• Choose Amazon Linux 2023.

• Configure:

- Default VPC
- Public Subnet
- Auto Assign Public IP

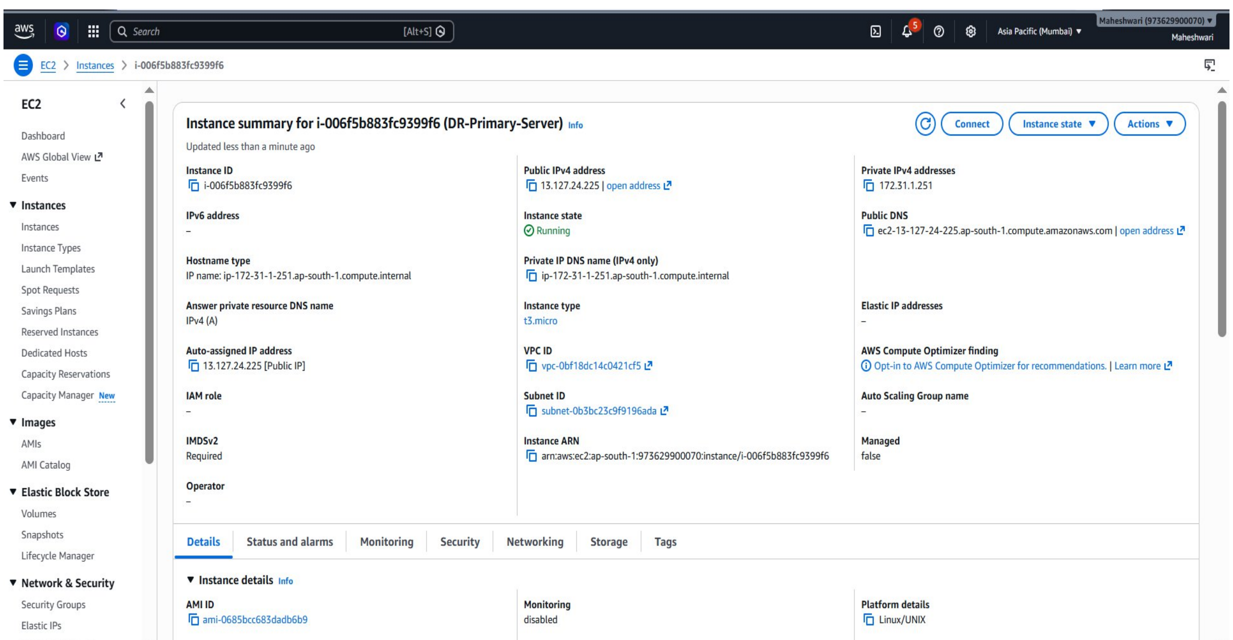
• Configure Security Group:

- SSH (22)
- HTTP (80)

• Launch Instance.

• Name:

DR-Primary-Server



Step 2: Install Web Server

Connect to EC2 using SSH.

Install Apache:

sudo yum update -y

sudo yum install httpd -y

sudo systemctl start httpd

sudo systemctl enable httpd

Create Web Page:

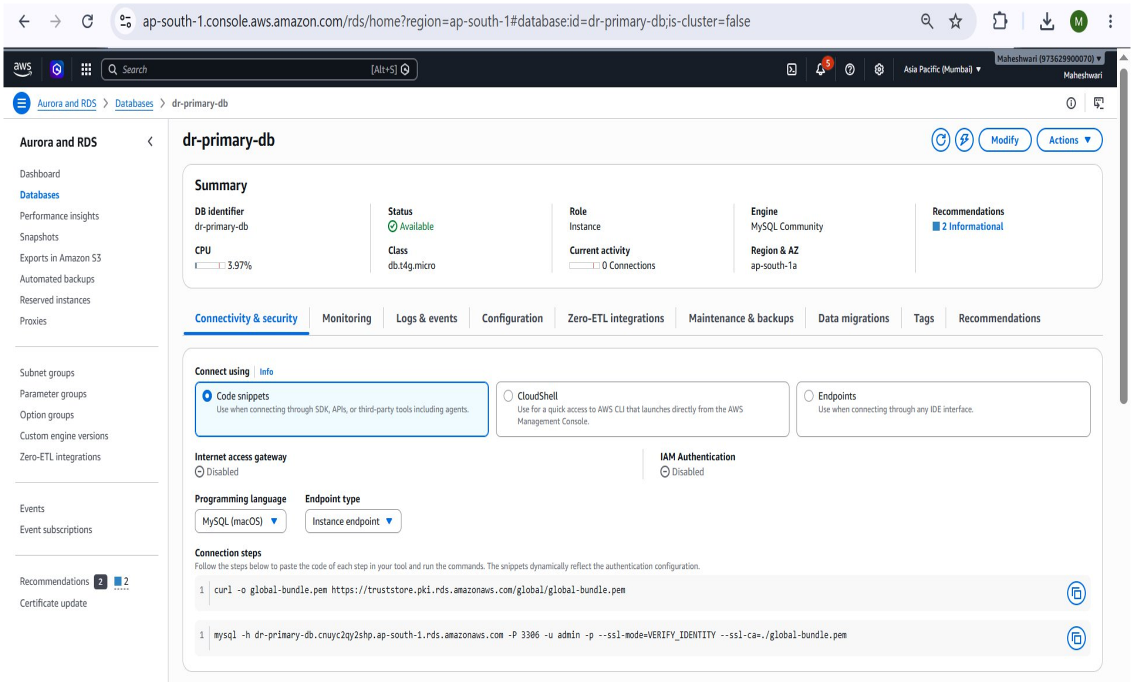
```
echo "Primary Region Website" | sudo tee /var/www/html/index.html
```

Verify Website:

http://Primary-EC2-Public-IP

Step 3: Create Primary RDS Database

- Go to AWS Console → RDS → Create Database.
- Engine: MySQL
- DB Identifier: dr-primary-db.
- Username: admin
- Storage:20 GB
- Enable Automated Backups.
- Create Database.

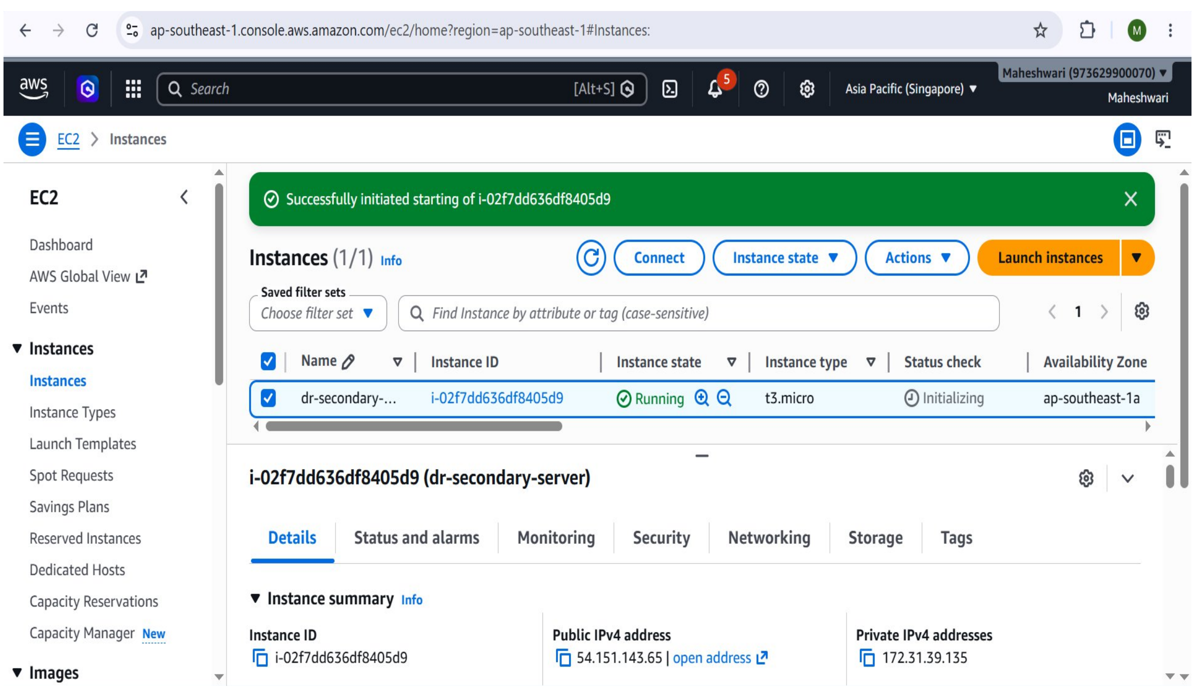


Step 4: Launch Disaster Recovery EC2 Instance

Switch Region: Singapore ap-southeast-1

Launch another EC2 Instance.

Name: DR-Secondary-Server



Install Apache.

Create Web Page:

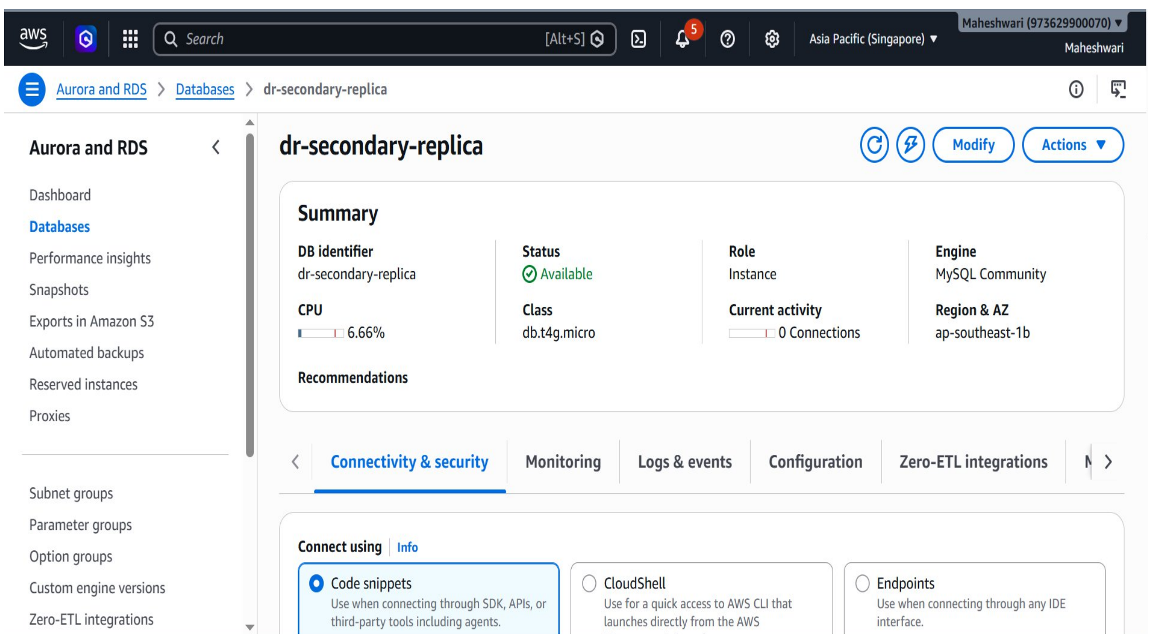
echo "Disaster Recovery Region Website" | sudo tee /var/www/html/index.html

Verify Website:

http://Secondary-EC2-Public-IP

Step 5: Configure Cross-Region Read Replica

- Open RDS Console.
- Select: Source-dr-primary-db.
- Destination Region = Singapore (ap-southeast-1)
- DB Identifier = dr-secondary-replica
- Actions → Create Read Replica.
- Destination Region: Singapore
- Create Replica.
- AWS creates: dr-secondary-replica

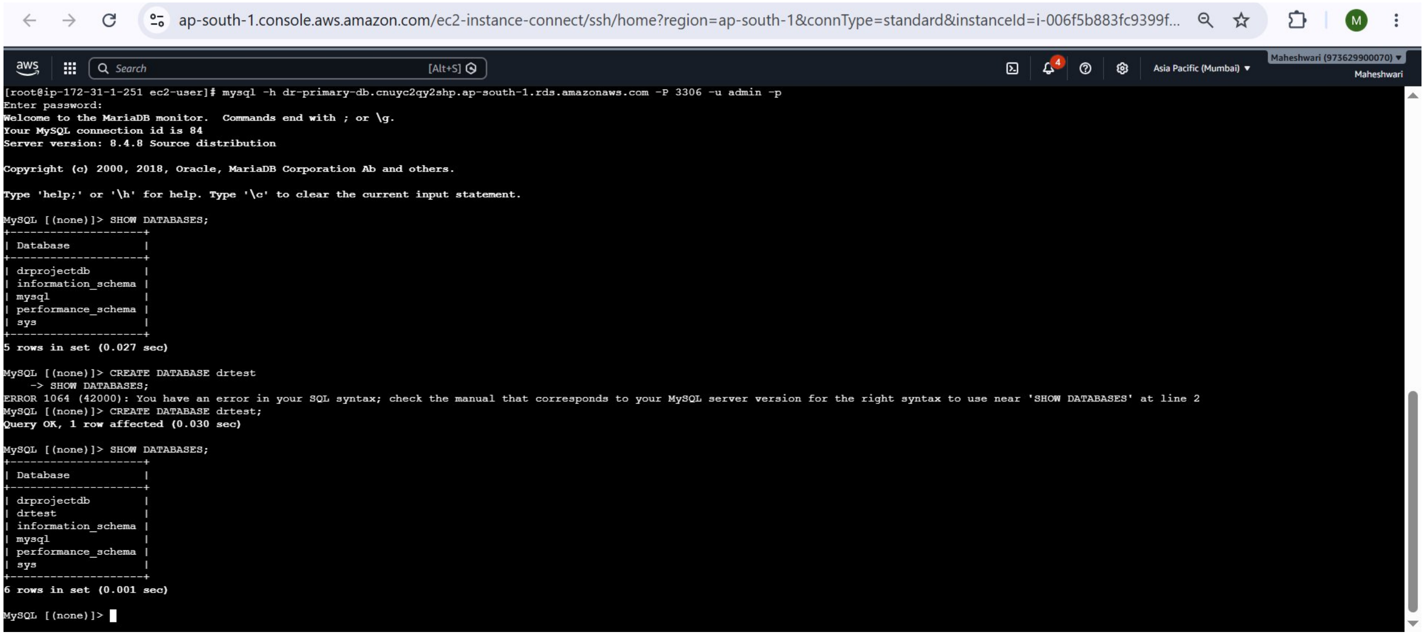


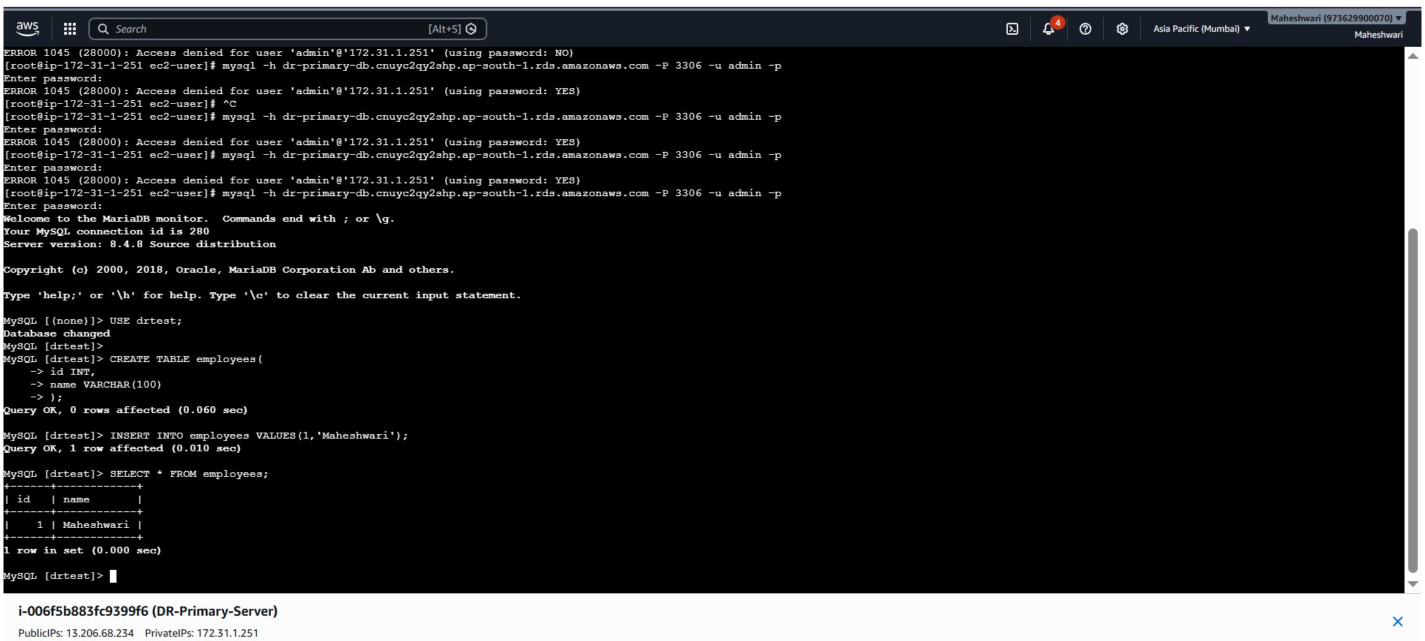
Step 6: Verify Database Replication

Connect to Primary Database.

Create Database:

CREATE DATABASE drtest;





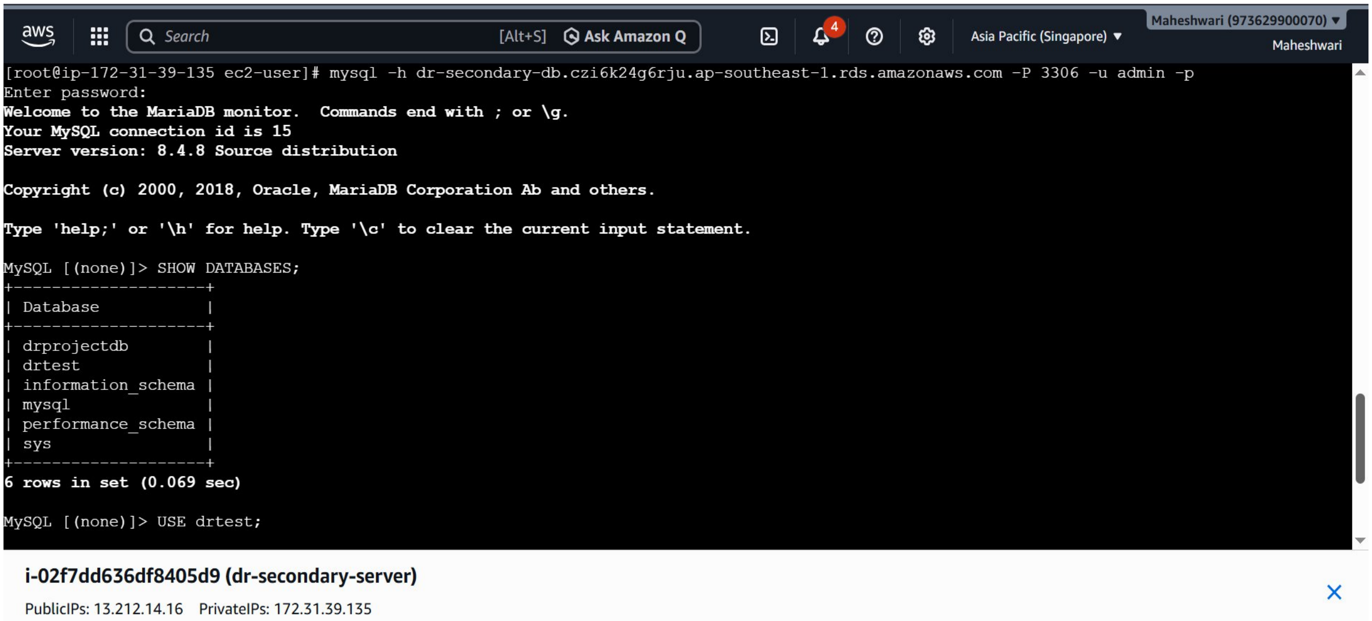
Connect to Replica Database.

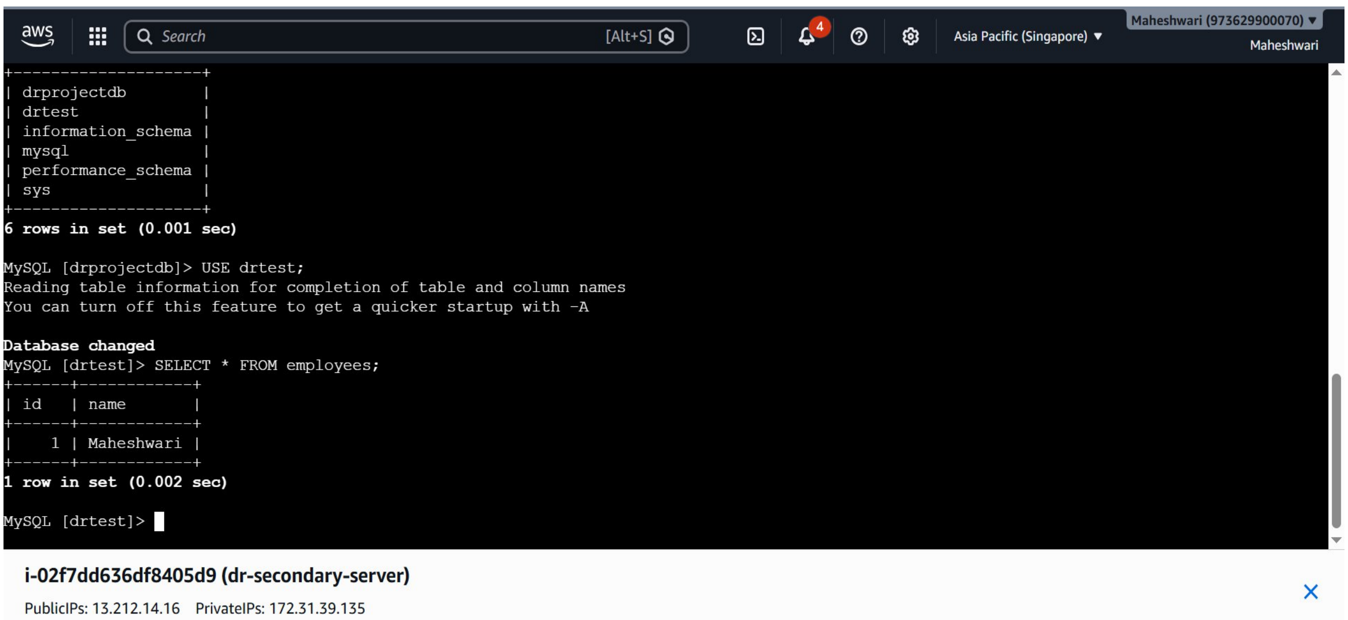
Run:

SHOW DATABASES

Verify: drtest

appears in the replica.





Step 7: Simulate Disaster Recovery

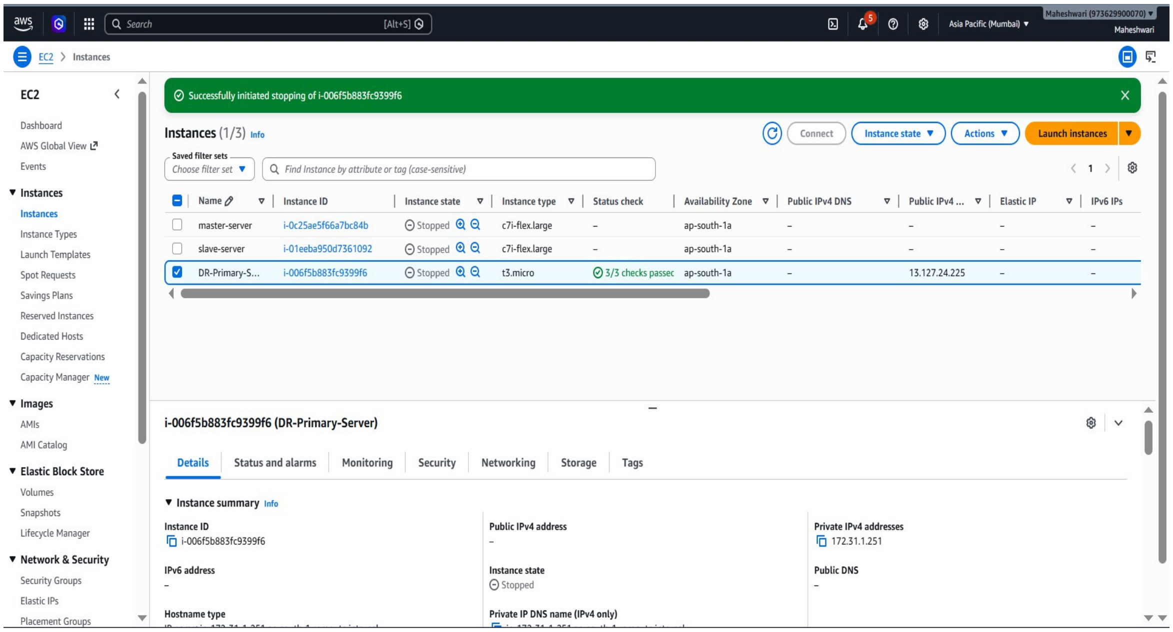
Stop the Primary EC2 Instance.

Go to:

EC2 → Instances

Select: DR-Primary-Server

Click: **Instance State → Stop**



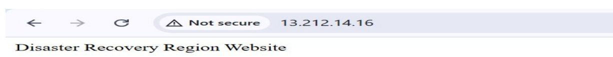
Step 8: Verify Disaster Recovery Environment

Access Secondary Region Website.

http://Secondary-EC2-Public-IP

Verify:

Disaster Recovery Region Website



Application remains available from the DR region.

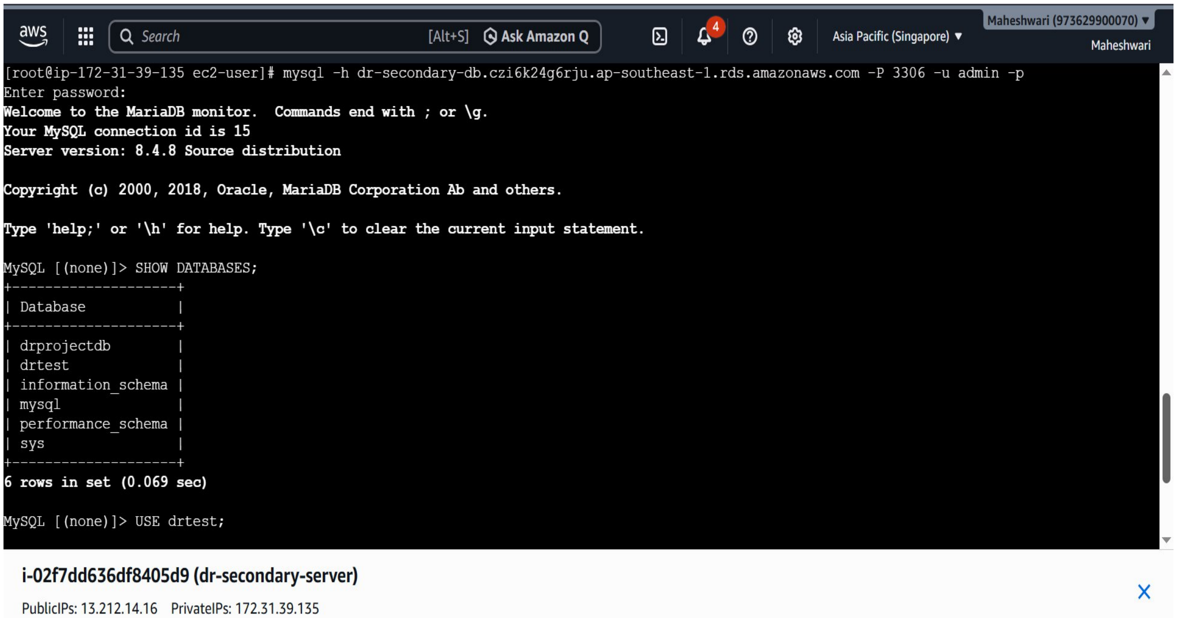
Step 9: Validate Database Availability

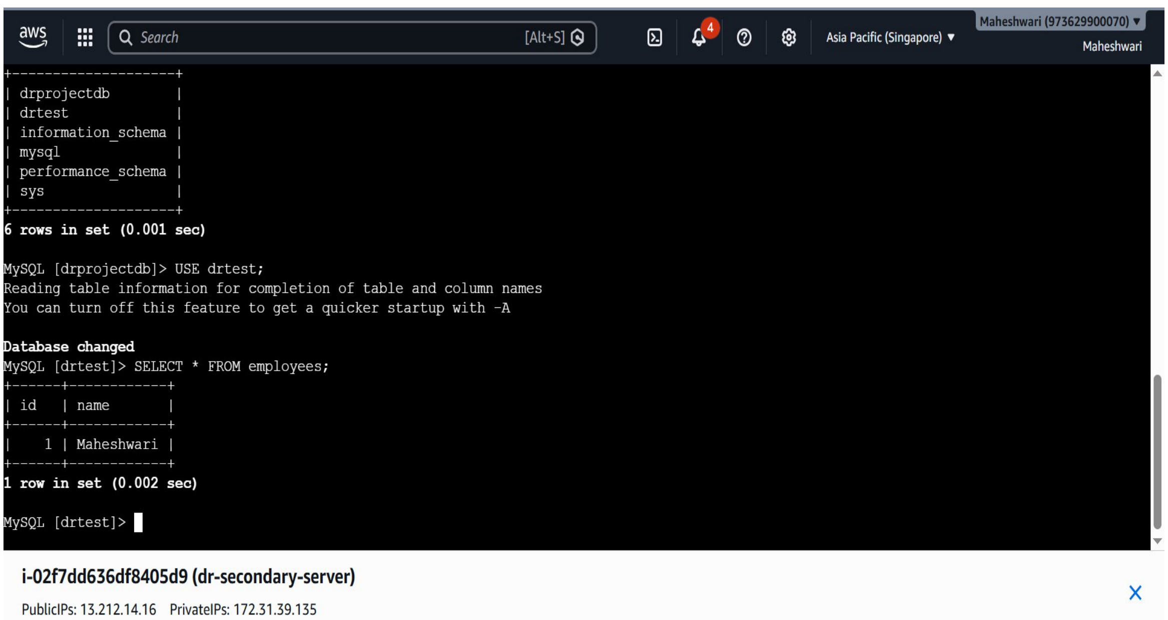
Verify:

• Replica database is available.

• Application can connect to the DR database.

• Recovery environment is operational.





Troubleshooting

Issue 1: Read Replica Creation Failed

The read replica may fail to create if automated backups are disabled.

Solution:

• Enable Automated Backups on the primary database.

• Retry Read Replica creation.

Issue 2: Replication Delay

Database changes may not immediately appear in the replica.

Solution:

• Wait a few minutes for synchronization.

• Verify replication status from the RDS console.

Issue 3: EC2 Website Not Accessible

The website may not load due to security group restrictions.

Solution:

• Verify HTTP Port 80 is allowed.

• Ensure Apache service is running.

sudo systemctl status httpd

Issue 4: Database Connection Failed

Application cannot connect to RDS.

Solution:

• Verify RDS security group.

• Check endpoint configuration.

• Confirm database credentials.

Conclusion

The Multi-Region Disaster Recovery Plan provides a reliable and scalable solution for ensuring business continuity across AWS regions. By deploying application servers in multiple regions and enabling Amazon RDS cross-region replication, organizations can minimize downtime and reduce the risk of data loss during regional failures.

The architecture improves availability, supports disaster recovery objectives, and demonstrates real-world cloud resilience practices. The solution is suitable for enterprise workloads requiring high availability, business continuity, and disaster preparedness.